



FACULTY
OF EDUCATION

Masaryk University

Vim (Presentation #4)

Special Topics (CS3001)

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A Brief Introduction

They often say

"A Developer Is Only As Good As His Tools"

Although I don't agree with this sentiment, I always wanted to become better with tools I was constantly using. One of the tools I use most is my code editor: **Vim**.

A Brief Introduction

- Vim has been around since 1991, and is almost ubiquitous on all Unix machines. It has served mostly as a terminal text editor (see attached video).
- Also, Vim has greatly improved a lot since then, to incorporate many of the new features that IDEs and Text Editors have brought to the table.
- Along with that, a lot of projects have set out to make Vim look and feel more like a regular text editor (as apposed to a terminal editor).

A Brief Introduction

```
matrix-base.hpp matrix-base.h
8 //
9 // matrix-base.h
10 // lib
11 //
12 // Il'ya Starikov/Rachel Milliam, Section 1A, 19883141
13 // cs2201 - Object Oriented Numerical Modeling
14 //
15
16 #ifndef matrix_base_h
17 #define matrix_base_h
18
19 #include "matrix.h"
20
21 // Forward Declaration For RectangularMatrix
22 template <class T>
23 class RectangularMatrix;
24
25 // Base Class BaseMatrix
26
27 // Brief A base that defines default behavior for a matrix.
28
29 // Signature T must have the following operations defined:
30 // - Addition (+)
31 // - Subtraction (-)
32 // - Multiplication (*)
33 // - Equal To (==)
34 // - Less Than (<)
35 // - Greater Than (>)
36 // - Negation (!)
37
38 template <class T>
39 class BaseMatrix : virtual public Matrix<T> {
40     static T _zero;
41
42 public:
43     template <class S> friend bool operator==(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
44     template <class S> friend bool operator!=(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
45
46     template <class S> friend RectangularMatrix<S> operator+(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
47     template <class S> friend RectangularMatrix<S> operator-(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
48     template <class S> friend RectangularMatrix<S> operator*(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
49     template <class S> friend RectangularMatrix<S> operator/(const BaseMatrix<S>& leftHandSide, const BaseMatrix<S>& rightHandSide);
50     template <class S> friend RectangularMatrix<S> operator+(const BaseMatrix<S>& leftHandSide, const Vector<S>& rightHandSide);
51     template <class S> friend RectangularMatrix<S> operator-(const BaseMatrix<S>& leftHandSide, const Vector<S>& rightHandSide);
52     template <class S> friend RectangularMatrix<S> operator*(const BaseMatrix<S>& leftHandSide, const Vector<S>& rightHandSide);
53     template <class S> friend RectangularMatrix<S> operator/(const BaseMatrix<S>& leftHandSide, const Vector<S>& rightHandSide);
54     template <class S> friend bool isDiagonallyDominant(const BaseMatrix<S>& leftHandSide);
55
56     virtual T& zero() const noexcept override;
57
58     template <class S> friend std::ostream& operator<<(std::ostream& os, const Matrix<S>& other);
59 };
60
61 #include "matrix-base.hpp"
62
63 #endif // matrix-base.h
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Prior Knowledge

- I've been using Vim since my freshmen year. However, I never worked with the advanced features of Vim, only the top level features.
- I knew that Vim had an expansive feature set, and I wanted to dive into that.

Motivation

- Upon graduation, I will likely be using Vim full time at my time at Garmin.

Goals

- To become more efficient at using Vim.
- To become familiar with the modern parts of Vim.
- To learn more of the keystrokes and shortcuts for Vim.

Resources

- A lot of the learning came from playing around with Vim. Anytime I found myself doing something repetitive, I simply tried to find a way to automate. Most of the time, there was a way.
- Book Resources (That I Actually Bought!):
 - [Modern Vim](#)
 - [Practical Vim](#)
- Online Resources:
 - [The Vim Subreddit](#), for everyday improvements or general knowledge on Vim.
 - [VimCasts](#), for when I want to get a deeper look into a subject.
 - [Vim Adventures](#), a little game to help understand Vim through a text adventure.

Goal Accomplishment

- I am much more fluent in using Vim, using shortcuts that significantly improve my workflow.
 - See attached videos for details.
- I have also learned to use plugins, to do completions.
- I have also learned to use many of the different parts of Vim I was unfamiliar with, such as:
 - Buffers
 - Registers
 - Advanced Search
 - Tags

And a lot more!

Side Note

Figure: The entirety of this presentation was made in Vim.

In Closing

All question, comments, and insults can be directed towards me:

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